

4) Límites infinitos:

$$a) \lim_{x \rightarrow \infty} \frac{2x+3}{3x+1}$$

$$= \lim_{x \rightarrow \infty} \frac{x(2 + \frac{3}{x})}{x(3 + \frac{1}{x})}$$

$$= \lim_{x \rightarrow \infty} \frac{2 + \frac{3}{x}}{3 + \frac{1}{x}}$$

$$= \frac{2+0}{3+0} = \boxed{\frac{2}{3}}$$

$$b) \lim_{x \rightarrow \infty} \frac{5x^2 + 3x + 1}{2x^2 - 4x - 5}$$

$$= \lim_{x \rightarrow \infty} \frac{x^2(5 + \frac{3}{x} + \frac{1}{x^2})}{x^2(2 - \frac{4}{x} - \frac{5}{x^2})}$$

$$= \lim_{x \rightarrow \infty} \frac{5 + \frac{3}{x} + \frac{1}{x^2}}{2 - \frac{4}{x} - \frac{5}{x^2}}$$

$$= \frac{5+0+0}{2-0-0} = \boxed{\frac{5}{2}}$$

$$e) \lim_{x \rightarrow \infty} \frac{4x^5 - 6x^4 + 3x^2}{3x^3 + 5x^2 + 6x}$$

$$= \lim_{x \rightarrow \infty} \frac{x^5(4 - \frac{6}{x} + \frac{3}{x^3})}{x^3(3 + \frac{5}{x} + \frac{6}{x^2})}$$

$$= \lim_{x \rightarrow \infty} \frac{x^2(4 - \frac{6}{x} + \frac{3}{x^3})}{3 + \frac{5}{x} + \frac{6}{x^2}}$$

$$= \infty \cdot \frac{4}{3} = \boxed{\infty}$$

$$f) \lim_{x \rightarrow \infty} \sqrt{\frac{1+x}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \sqrt{\frac{x(1 + \frac{1}{x})}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \sqrt{\frac{1 + \frac{1}{x}}{x}}$$

$$= \sqrt{\frac{1+0}{\infty}} = \sqrt{0} = \boxed{0}$$

$$g) \lim_{x \rightarrow \infty} \sqrt{3x+2} - x$$

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{3x+2} - x)(\sqrt{3x+2} + x)}{\sqrt{3x+2} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{3x+2 - x^2}{\sqrt{3x+2} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{-x^2 + 3x + 2}{x + \sqrt{3x+2}}$$

$$= \frac{-\infty}{\infty} \sim \frac{-x^2}{x} = \boxed{-\infty}$$

$$h) \lim_{t \rightarrow \infty} \frac{3t^4 + 3t^3 + 3t}{4t^4 + 2t^3}$$

$$= \lim_{t \rightarrow \infty} \frac{t^4(3 + \frac{3}{t} + \frac{3}{t^3})}{t^4(4 + \frac{2}{t})}$$

$$= \lim_{t \rightarrow \infty} \frac{3 + \frac{3}{t} + \frac{3}{t^3}}{4 + \frac{2}{t}}$$

$$= \frac{3+0+0}{4+0} = \boxed{\frac{3}{4}}$$